

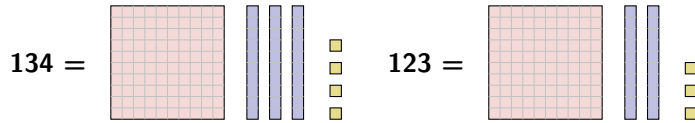
Introduction to SG Math

Base-Ten Place Value Blocks & Bar Models

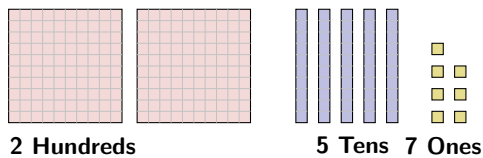
Part 1: Base-10 Block Addition

Example: Solve $134 + 123$ using Hundreds (Flats), Tens (Rods), and Ones (Units).

① Represent each number using blocks



② Group hundreds, tens, and ones together

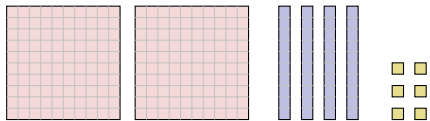


③ Write the final value

$$2 \text{ Hundreds} + 3 \text{ Tens} + 7 \text{ Ones} = 200 + 30 + 7 = \mathbf{237}$$

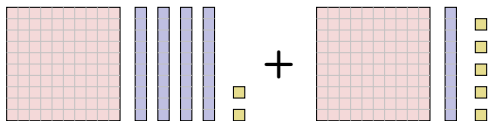
Practice Questions

1. What total number is shown by the blocks below?



_____ Hundreds + _____ Tens + _____ Ones = _____

2. Add the blocks to find $142 + 115$:



Total = _____ Hundreds + _____ Tens + _____ Ones = _____

Part 2: Base-10 Subtraction (Taking Away)

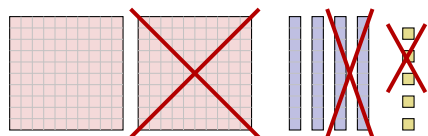
Example: Solve $245 - 123$ by crossing out blocks.

① **Build the starting number (245)**

Start with 2 Hundreds, 4 Tens, and 5 Ones.

② **Take away 123 (1 Hundred, 2 Tens, 3 Ones)**

Cross out the subtracted amounts with red lines:



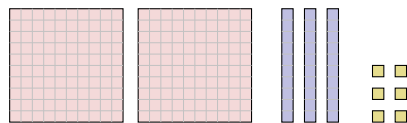
③ **Count remaining blocks**

Remaining = 1 Hundred, 2 Tens, 2 Ones

$$245 - 123 = 122$$

Practice Questions

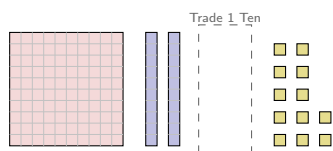
1. **Cross out blocks to solve** $236 - 114$:



Remaining: _____ Hundreds + _____ Tens + _____ Ones = _____

2. **Regrouping Tens to Ones:** $132 - 115$

Hint: Trade 1 Ten Rod for 10 Unit Cubes first, then take away 5 ones.



Equation: $132 - 115 =$ _____

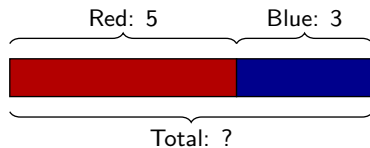
Part 3: Bar Models

Example: Maya has **5 red pencils** and **3 blue pencils**. How many pencils in total?

① **Identify the parts**

- Red pencils = 5
- Blue pencils = 3

② **Draw the bars**

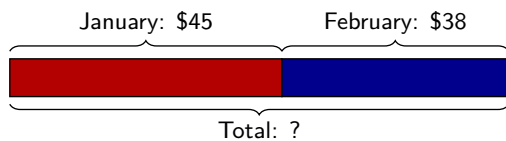


③ **Write and solve**

$$5 + 3 = 8$$

Practice Questions

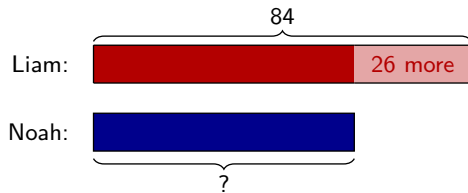
1. Lucas saved \$45 in January and \$38 in February. How much in total?



Equation: _____ + _____ = _____

Total = \$ _____

2. Liam has 84 stamps, 26 more than Noah. How many does Noah have?



Equation: _____ - _____ = _____

3. Apples : oranges is 4 : 7. There are 28 apples. How many oranges?

Apples: 

Oranges: 

1 unit = $28 \div 4 =$ _____

Oranges = $7 \times$ _____ = _____